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A Smart Rehabilitation Glove Based on Shape-Memory Alloys for Stroke Recovery

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Abstract

Stroke-induced hand dysfunction substantially impairs patients' quality of life, creating an urgent need for portable, adaptive rehabilitation devices. This study introduces a smart rehabilitation glove actuated by shape-memory alloy (SMA) wires, leveraging their high power-to-weight ratio, controllable strain recovery, and reversible phase transformation to overcome the limitations of conventional motor-driven or pneumatic gloves. The glove incorporates SMA-based actuation units achieving 50 mm contraction (5% strain) within 7 s, enabling finger flexion to ~34° for personalized rehabilitation protocols. A mobile application provides wireless regulation of SMA actuation modes and facilitates real-time telemedicine consultations. The prototype demonstrates an ultra-lightweight, compact design enabled by SMA's intrinsic properties, offering a promising solution for home-based post-stroke rehabilitation. This work establishes the transformative potential of SMAs in wearable biomedical technologies.

Keywords: shape-memory alloys; rehabilitation glove; smart wearable device; stroke recovery

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1. Introduction

Stroke is an acute cerebrovascular disease that damages the brain tissue due to the sudden rupture of a blood vessel in the brain or the failure of blood to flow towards the brain. Studies show that one in four people worldwide will experience a stroke, which is the second leading cause of death globally, with the number of cases continuing to rise [1–3]. Patients after stroke usually experience hand disorders, and neuroplasticity theory suggests that repetitive and consistent hand recovery exercises have the potential to reconnect the neural pathways that regulate hand function, which are effective in treating stroke patients [4,5]. Hence, over the past decade, scientists have paid attention to rehabilitation

training devices designed to treat hand dysfunction, demonstrating that repetitive rehabilitation training has ability to enhance hand motor function with the assistance of robots [6,7]. However, traditional rehabilitation methods are time-consuming, and the frequency and intensity of recovery cannot be well controlled [8–10]. Nowadays, exoskeleton-assisted rehabilitation machines effectively help patients to repeat hand movements periodically to complete therapy [11].

Currently, rehabilitation gloves are mainly divided into pneumatic and motor-driven types [12–15]. A micro-motor-driven rehabilitation glove developed by Chen et al. leverages machine learning to recognize the gestures of the sensing glove [16]. However, the glove driven by the motor is bulky. Additionally, a hydraulically actuated rehabilitation glove designed by Polygerinos et al. uses soft, segmented elastic actuators reinforced with fiber materials, facilitating specific bending, twisting, and stretching motions. Although it is effective for achieving sports rehabilitation, the pneumatic-driven glove is not easy to control, and the gas is difficult to seal [17].

In contrast to conventional approaches, shape-memory alloys (SMAs) have emerged as a significant research focus for rehabilitation exoskeleton devices due to their unique intelligent characteristics. The most remarkable feature of these materials is the shape-memory effect (SME), enabling recovery of their preset configuration under specific stimuli (e.g., mechanical stress or thermal activation). This behavior originates from a reversible martensitic transformation (MT) between an austenite phase (B2 cubic structure) at high temperature and a martensite phase (B19′ monoclinic structure) at low temperature. Cooling below the martensite finish temperature (M_f) induces the forward transformation (B2→B19′), allowing deformability at low stresses. Subsequent heating above the austenite start temperature (A_s) triggers the reverse transformation (B19′→B2), recovering the original shape [18–24]. When electrified, SMA wires generate Joule heating that elevates the temperature above A_s, driving the martensite-to-austenite transformation. This phase change produces characteristic "low-temperature compliance and high-temperature contraction" behavior. The B2-to-B19′ notation explicitly references the crystallographic restructuring during MT, which underlies the actuation mechanism in this study. Some studies have shown the feasibility of SMA for use in rehabilitation devices [6,25–29]. For example, Kauffman et al. developed a wearable hand-rehabilitation robotic arm, driven by SMA wires with corresponding joints equipped with sensors on the glove, to fix the path of SMA contraction [18]. The force generated by SMA contraction is used to compensate for insufficient finger-muscle force. Additionally, a variable-stiffness robot finger has been developed where the finger is driven by three embedded SMA wires [19]. Additionally, our research team previously filed a patent (CN 216,908,523 U) for an SMA-based rehabilitation glove, which laid the foundation for the amplified actuation structure described in this study [30]. Currently, commonly used SMAs include NiTi and NiTi-based alloys, Cu-based shape-memory alloys, and Fe-based shape-memory alloys. Among these, near-equiatomic NiTi alloys are widely used for their excellent mechanical properties and fatigue performance. But the recoverable strain of NiTi alloys is typically less than 8%; thus, a long SMA wire is required to provide sufficient contraction length for recovery treatment, which increases the volume of devices [20–24].

This study proposes a lightweight, small-volume, and portable rehabilitation glove based on SMA wires, facilitating repetitive recovery motion of disabled fingers for stroke treatment. This smart rehabilitation glove is equipped with an amplifier that amplifies the stroke of the glove six times through multiple windings of SMA wires, capable of accommodating 960 mm of wires within a 160 mm amplifier. This increases the stroke from 8.3 mm to 50.2 mm, providing sufficient displacement for finger recovery. Furthermore, the wireless-transfer modules are assembled on the glove and a mobile application is

designed to achieve online health monitoring for patients and doctors, greatly improving the convenience of diagnosis and treatment. In this work, we present an SMA-actuated rehabilitation glove with a specific amplifier structure and control mode, and verify the performance of the product through experiments.

2. Design

In this design, SMA wires are placed inside the glove to generate driving force under currents, which enables finger motion to be achieved without using an electric motor or air pump. After heating up with electricity, the SMA wires assembled on the amplifier undergo phase transformation and contract, driving the carbon wires to complete the finger rehabilitation exercise. Meanwhile, by adjusting the current intensity and frequency, the training intensity and speed can be changed. For example, when selecting a high-intensity mode, a high current value is output, causing the SMA wires to contract faster, enabling finger movement to reach a certain degree of bending in a shorter time to accommodate the patient's condition. Furthermore, the mobile app enables real-time monitoring of training data throughout the recovery process, facilitating feedback regarding the patient's condition and the doctor's diagnosis. The experiments in this study show that the smart rehabilitation glove prototype is convenient to wear, suitable for hands of different sizes, capable of providing rehabilitation training in different modes, and capable of realizing medical monitoring.

2.1. Structural Design

The overall components of the smart rehabilitation glove are shown in Figure 1a. The glove mainly consists of SMA wires, carbon wires, an amplifier, an aluminum shell, and an outer shell. Within the aluminum shell, each carbon wire is connected with a corresponding SMA wire to pull each finger for recovery movements. To complete rehabilitation training, a contraction displacement of 54 mm is required for each finger. The SMA can provide, at most, 8% recoverable strain; thus, a total of at least 680 mm of SMA wires is needed, which takes up a large volume in the glove [31]. To minimize the size of glove, an amplifier structure was designed, as shown in Figure 1b, taking up a small volume and providing an amplified contraction. The lines represent the SMA wires assembled on insulated shafts, and the pulley wheel represents the fixed pillar to keep the carbon wires and SMA tight and stable. Furthermore, due to the heat generated by the SMA wires when powered on, the carbon wires are connected to the SMA wires in the finger region for user safety. One side of the SMA is fixed, while the other side is continuously winded around the pulley wheel to achieve reciprocating coiling, resulting in an N-fold amplification of the contraction (N is the number of coiling times), effectively reducing the space occupied by the SMA wires and the volume of product. In the glove, a single finger employs a value of N = 6, corresponding to a 6-fold amplification, which enables the accommodation of an SMA with a length of $160 \times 6 = 960$ mm, thus fulfilling the stroke requirements for each finger. Using the amplification system, multiple SMA wires and shafts are combined to form the force-driving module that is connected to the finger joints to become the prototype of a smart rehabilitation glove, enabling the recovery movement of fingers from (I) a natural state to (III) a fully extended state, as shown in Figure 1c.

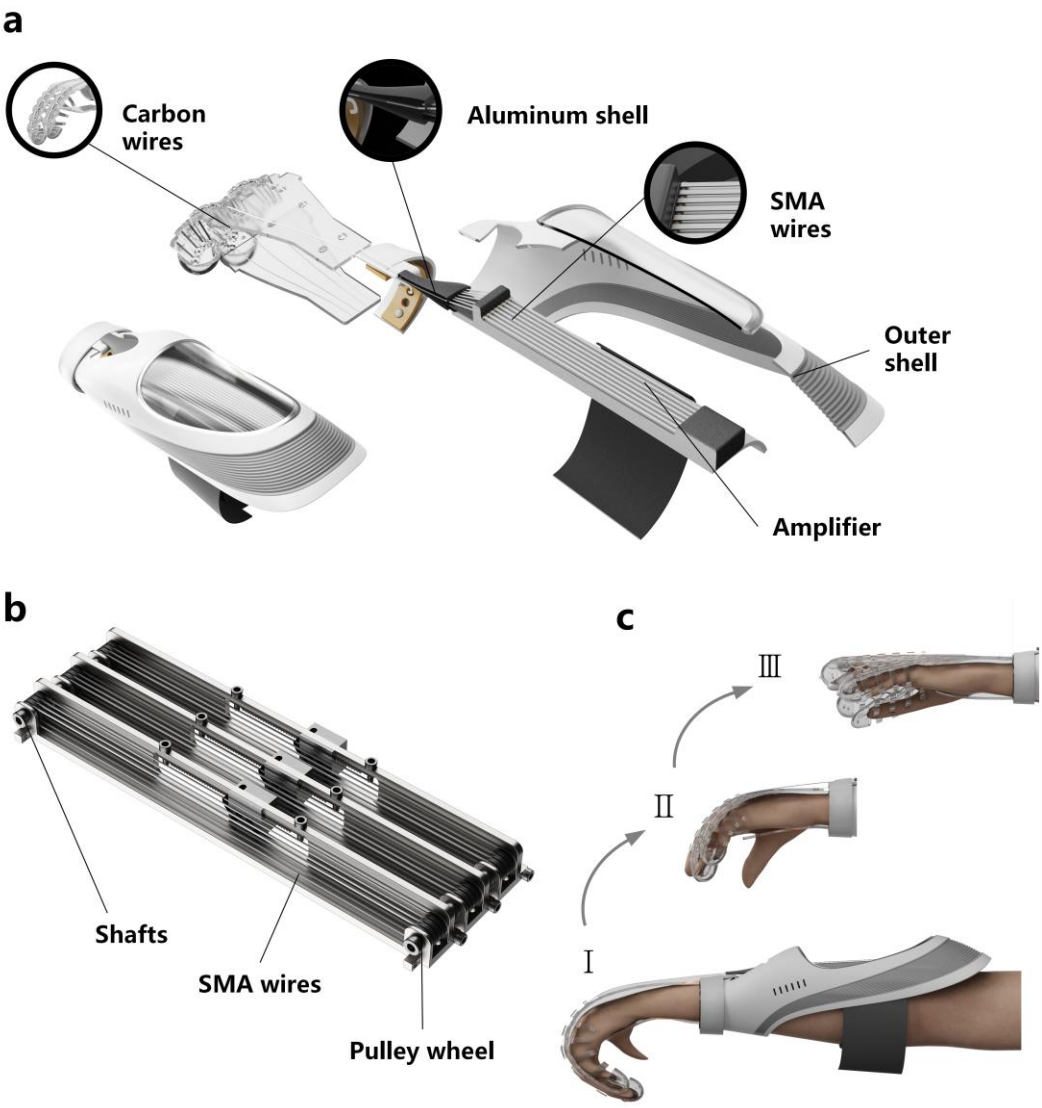

Figure 1. A schematic diagram of the smart rehabilitation glove. (**a**) The structural components of the rehabilitation glove. (**b**) Details of the amplifier. (**c**) The recovery process of a dysfunctional hand, including (I) the natural state, (II) the intermediate state, and (III) the fully extended state.

2.2. Online Control in Mobile App

As shown in Figure 2, the online real-time monitoring function of the rehabilitation glove provides an efficient and low-cost health assessment solution for patients with hand dysfunction and their doctors. Among them, Figure 2a presents the perspective of the patient-side interface, and Figure 2b shows the perspective of the doctor-side interface. The mobile application employs a Flutter-based three-tier architecture:

Frontend: Patient interface with mode sliders (low/med/high) and training dashboards; doctor interface for multi-patient real-time monitoring.

Communication: BLE 5.0 enables 128 kbps encrypted transmission (AES-128, implemented by Nordic Semiconductor, Oslo, Norway), with an end-to-end latency of 120 ms.

Cloud: Firebase stores anonymized data (SHA-256 encryption, provided by Google LLC, Mountain View, CA, USA) for remote telemedicine syncing.

For elderly users, the UI features high-contrast color blocks with <2 s anomaly alert response (ISO 13482 compliant [32]). To further address potential accessibility challenges, the system integrates two additional components: (1) a physical tactile button module for direct activation of the rehabilitation glove, enabling operation without a smartphone;

(2) a caregiver-assist mode that supports real-time monitoring and remote management by family members. These multi-faceted measures collectively enhance technological accessibility for elderly users, aligning with best practices in gerontechnology design and ensuring seamless integration of rehabilitation therapy into daily life.

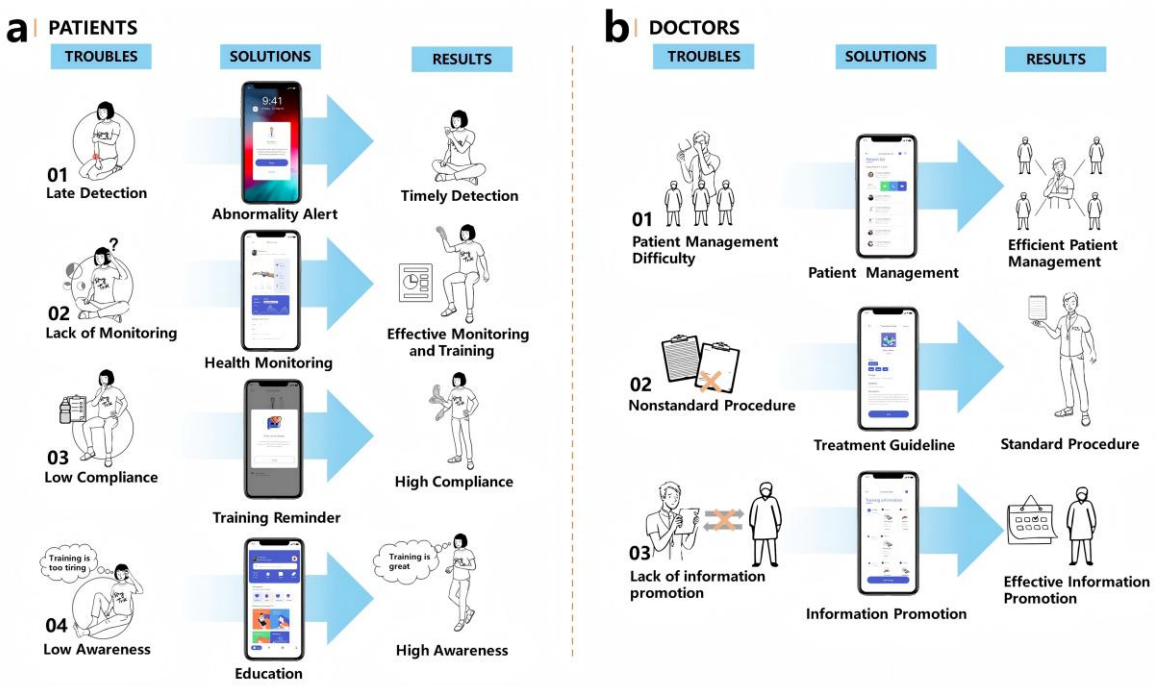

Figure 2. Online functions in mobile application for patients (**a**) and doctors (**b**), respectively.

A data transmission workstation is used to collect signals from patients when using the application. Sensors embedded in the rehabilitation gloves gather various physiological signals that are subsequently processed by the chip, converted into electrical signals, and transmitted to the application via a wireless-transfer module [33]. The data collected on the mobile is constantly uploaded to the cloud server for storage through the network. Meanwhile, the application and cloud server are responsible for the generation, transmission, and storage of the data. Overall, the use of an SMA-based rehabilitation glove in conjunction with an online monitoring application offers several advantages. Firstly, this type of glove is user-friendly to wear, due to the compact and lightweight design of the product. Patients are allowed to select a proper mode to complete their daily training based on their specific condition via the mobile application, which is especially convenient for elderly patients. Moreover, doctors are capable of remotely tracking the patient's recovery progress and receiving timely feedback, which greatly reduces healthcare costs.

3. Method

3.1. SMA Materials and Methods

To perform rehabilitation exercises for patients with different conditions, the diameter and length of SMA wires were selected to provide sufficient driving force and contraction length for training. The basic movements of rehabilitation training involve the bending and straightening of the finger joints. The minimum motion requirement for patients in the early stage of rehabilitation is the achievement of a finger joint circumference C value of 68 mm. We selected a $Ni_{50.3}Ti_{49.7}$ SMA wire with a diameter of 0.15 mm, which was commercially obtained from PeierTech, Jiangyin, China. The mechanical characteristics of the SMA wires approached a steady state after thermomechanical pretraining. The prestress was 500 MPa during pretraining, and 60 cycles of Joule heating at 40 mA was performed,

which induced saturated dislocations in the NiTi wires to suppress the accumulation of defects in the subsequent cyclic loading fatigue process, and limited the generation of the R phase. The pretraining process helped to stabilize the performance and improved both functional and structural fatigue [34].

The mechanical properties of both the SMA wires and the fabricated prototype were systematically characterized. Uniaxial tensile tests were performed by dynamic mechanical analysis (DMA 850, TA Instruments, New Castle, DE, USA) with a temperature control box. All DMA tests were conducted under uniaxial tension at constant strain rates, without frequency sweep. The sample was loaded at a strain rate of $2.5 \times 10^{-3}\,s^{-1}$. Some samples were strained to failure, whereas others were strained to 3%, 5%, and 7% strain.

The thermomechanical tests were conducted at constant temperatures at −15 °C, 15 °C, 45 °C, and 75 °C, with a controlled strain of 5% and a strain rate of $5 \times 10^{-4}\,s^{-1}$, to obtain the corresponding stress–strain curves.

Strain–temperature curves were obtained by applying constant tensile stresses (150 MPa, 200 MPa, 300 MPa) with a cooling/heating rate of 5 °C/min.

The phase-transformation temperatures were determined by differential scanning calorimetry (DSC 8500, PerkinElmer, Waltham, MA, USA) measurements. The temperature range was −100 °C to 100 °C, with a cooling/heating rate of 10 °C/min, and 3 min isothermal holding at both 100 °C and −100 °C to ensure complete phase transformation.

The SMA wire was powered by a laboratory power supply, which generated Joule heating to increase the temperature of the SMA wire to its phase-transformation temperature. This setup was employed to investigate the relationship between the strain of the SMA wire and the magnitude of the applied current, thereby obtaining the current–strain characteristic curve.

In the experiment comparing contraction forces under different applied voltages, an SMA wire with a length of 15 cm and a diameter of 0.15 mm was secured at one end to a spring dynamometer, while the other end was attached to a weight to ensure the wire remained taut. The wire was subjected to different voltage inputs, and the variations in the dynamometer readings during the electrification process were recorded. Similarly, in the experiment investigating the influence of the length of the wire during electrification, a constant voltage of 4 V was applied and the variations in the readings of the spring dynamometer during the process were measured and recorded.

3.2. Wireless Connection of Mobile and Rehabilitation Glove

In the prototype developed in this study, the integrated circuit board embedded in the smart rehabilitation glove has the capability to collect, process, and transmit data. The sensors and wireless-transmission module within the board transmit the detected physiological signals to the mobile application, allowing patients and doctors to remotely obtain these parameters. The sensors include a piezoresistive pressure sensor and infrared thermometers (with an uncertainty of ±2%). The wireless-transmission module was selected for its stability, low-power consumption, and rapid response time in short-range communication [35]. This technology is widely employed in products that integrate wearable devices with mobile terminals.

To facilitate the usage of the application by both patients and doctors, the software Figma Online (Version 2025.05.15, https://www.figma.com, accessed 15 May 2025) was used to design the mobile application's interface, due to its various functions, including multi-user collaboration, interface interaction, and online design [36]. To help doctors assess the health of patients and adjust training modes, an interface called "patient health monitoring" is included in the doctor-specific version of the application. This interface contains physiological signals received and transmitted by sensors, as well as basic information

for the corresponding patient and physical analysis reports. When doctors receive data remotely, they can judge the patient's rehabilitation based on the data and adjust the rehabilitation training mode in a timely manner. This design makes communication between patients and doctors more convenient and cost-effective, benefiting the patient's recovery.

4. Results and Discussion

4.1. SMA Properties

The smart rehabilitation glove uses SMA wires as the actuation part, and the performance of the SMA wires is shown in Figure 3. Figure 3a shows the stress–strain curve of the SMA wires under tensile testing until fracture. The curve exhibits the characteristic mechanical behavior of SMAs, including an initial elastic deformation stage, followed by a stress plateau region associated with stress-induced martensitic transformation, and a subsequent stress-hardening stage dominated by plastic deformation. The material reaches its maximum load-bearing capacity at approximately 1600 MPa, before undergoing fracture at a strain of around 16%. These results indicate that the SMA undergoes significant stress-induced phase transformation during tension, and ultimately fails at a high-stress level, which guarantees that the SMA wires can provide a stable and sufficient driving force for the rehabilitation glove.

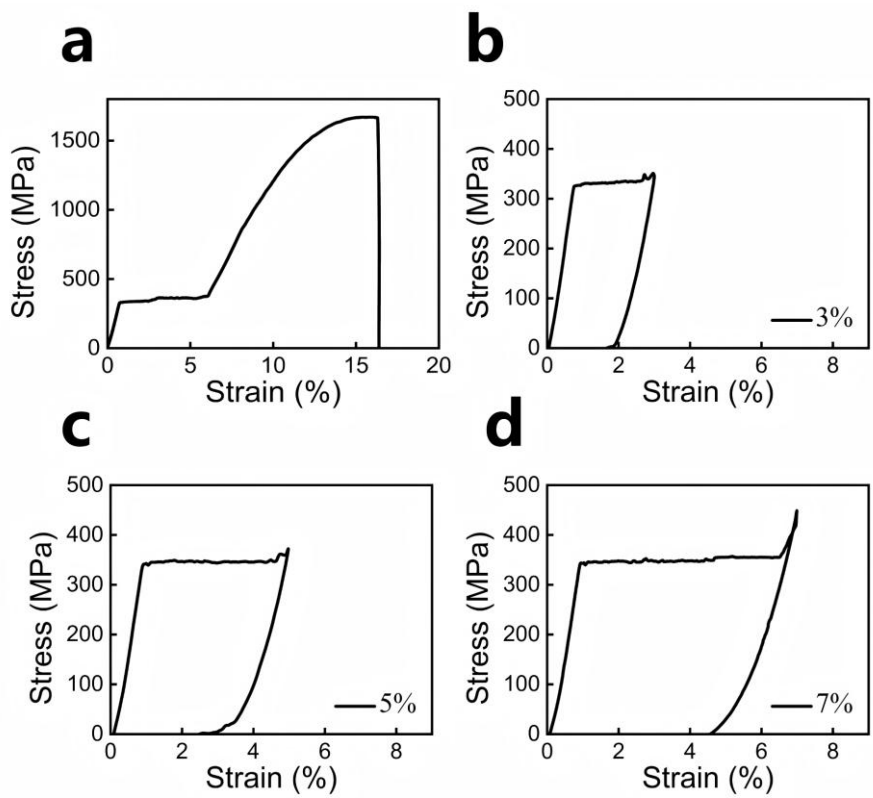

Figure 3. (**a**) The stress–strain curve of SMA wires under tensile testing until fracture. (**b**–**d**) The stress–strain curves of SMA wires under different strains of 3%, 5%, and 7%.

In Figure 3b–d, the stress–strain curves of SMA wires under different applied strains (3%, 5%, and 7%) exhibit characteristic pseudoelastic behavior. When the applied stress reaches the critical phase-transformation stress, a stress plateau appears due to the stress-induced B2→B19′ martensitic transformation. The critical stress for phase transformation (σ_c) is approximately 350 MPa. As the applied strain increases, the hysteresis loop becomes more pronounced, indicating enhanced energy dissipation. Upon unloading, most of the strain is recovered, with a residual strain that can be fully restored through heating.

Figure 4a illustrates the phase transformation of the SMA induced by temperature. During the cooling process, a single exothermic peak is observed, corresponding to the phase transformation from B2 to B19′ martensite. Conversely, the heating process exhibits a prominent endothermic peak, indicating the reverse transformation from the martensitic phase to the austenitic phase. The phase-transformation temperatures were determined using the double-tangent method, yielding A_s = 35.1 °C, A_f = 46.5 °C, M_s = 24.1 °C, and M_f = 13.4 °C. Here, A_s and A_f denote the start and finish temperatures of the austenitic transformation, whereas M_s and M_f correspond to the start and finish temperatures of the martensitic transformation.

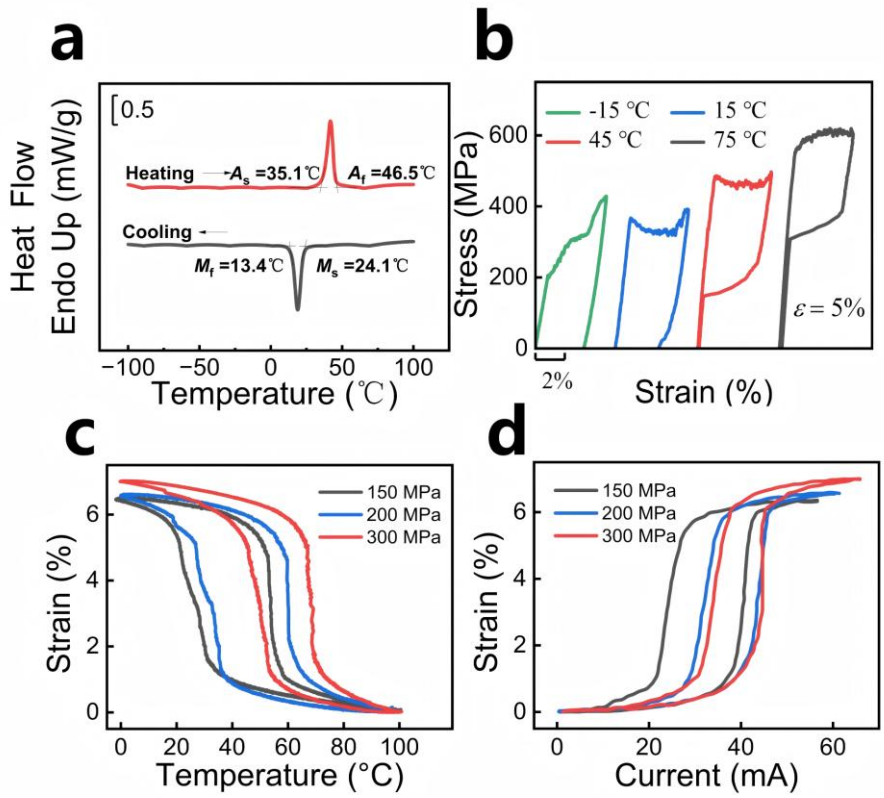

Figure 4. (**a**) The crystal structure of the SMA changes with temperature. (**b**) The stress–strain curves of SMA wires under 5% strain at selected temperatures (−15 °C, 15 °C, 45 °C, and 75 °C). (**c**) The temperature–strain curves of SMA wires under different stresses of 150 MPa, 200 MPa, and 300 MPa. (**d**) The current–strain curves of SMA wires under different stresses of 150 MPa, 200 MPa, and 300 MPa.

Figure 4b shows the stress–strain curves of the SMA wires at a fixed strain of 5% under different temperatures (−15 °C, 15 °C, 45 °C, and 75 °C). As the temperature increases, the plateau stress rises, indicating that higher critical stress is required for phase transformation. These results highlight the temperature dependence of the pseudoelastic behavior of SMAs.

The temperature–strain curves of the SMA wires under different stresses are shown in Figure 4c. As the stress increases, the phase-transformation temperatures of the SMA wires increase accordingly. The phase-transformation temperatures at 300 MPa were determined by the double-tangent method to be A_s = 62.1 °C, A_f = 72.3 °C, M_s = 55.6 °C, and M_f = 43.2 °C. As shown in Figure 4c, the recovery strain is about 6%.

SMA wire actuation tests were conducted at stresses of 150 MPa, 200 MPa, and 300 MPa (Figure 4d), with incrementally increasing current values from 0 to 60 mA and back to 0 mA. The experiments reveal that under 150 MPa stress, applying approximately 20 mA allows the temperature of the SMA wire to reach the phase-transformation value, initiating contraction. As the current increases, the amount of SMA strain increases. Moreover, as

the applied stress increases, the phase-transformation temperature also increases; thus, a current of about 30 mA is required to initiate phase transformation for SMA wires under 300 MPa.

As shown in Figure 5a, a fixed 0.15 mm diameter SMA wire with a length of 15 cm is fixed to a spring dynamometer at one end and is attached to a weight at the other end. A voltage (from 2 V to 8 V) is applied to the wire, and the results show that the time taken to achieve the same contraction force is reduced at a higher voltage, while the final contraction force remains nearly constant. However, when the applied voltage is too low (e.g., 2 V), the wire fails to reach the same contraction force. Figure 5b indicates that under a constant applied voltage of 4 V, an increase in SMA wire length leads to a corresponding increase in contraction force. Additionally, the time required to reach the final contraction state extends with increasing wire length. The experimental results demonstrate that selecting an appropriate voltage and SMA wire length can optimize the performance of the rehabilitation glove, ensuring adequate assistive force and responsive speed.

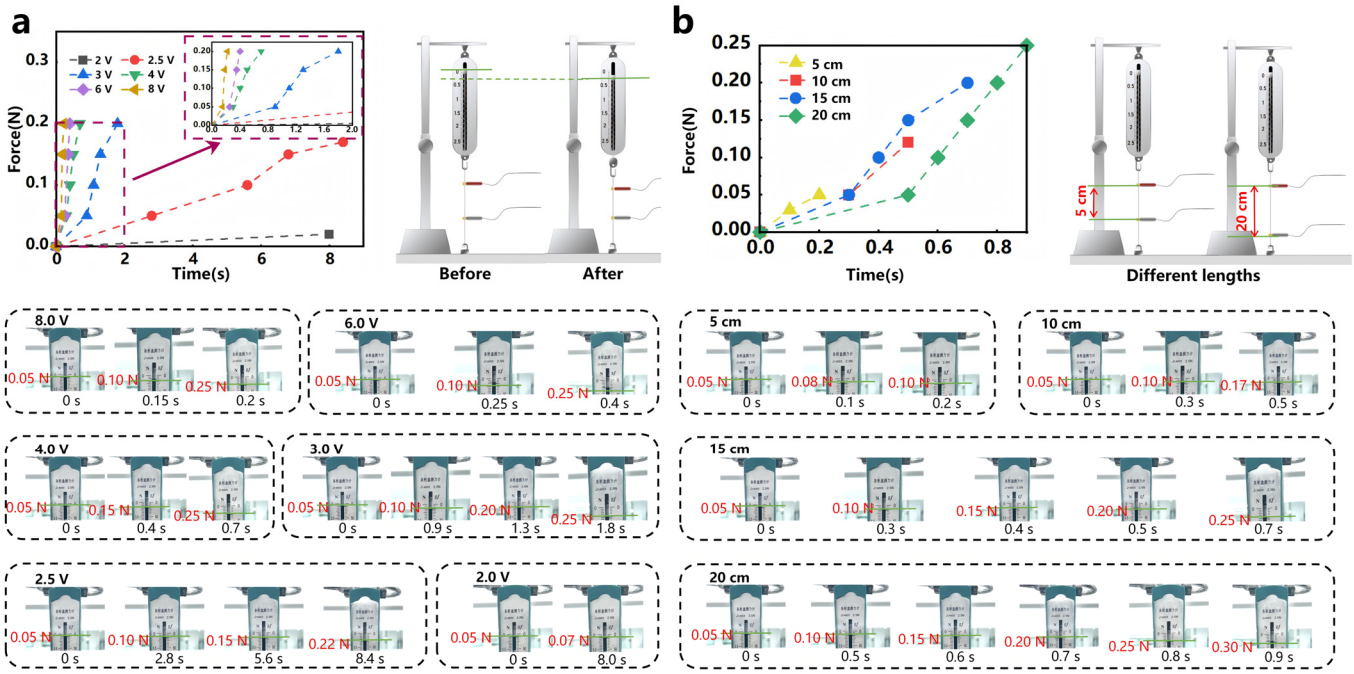

Figure 5. (**a**) The influence of different voltages on the output force. (**b**) The influence of different lengths on the output force.

4.2. Training-Mode Modulation

To enable patients to adjust the training modes according to their recovery process, a function for training-mode adjustment was developed for the mobile application. There are three buttons that patients can choose to switch the training modes: low, middle, and high, as shown in Figure 6a. When patients operate the application on a mobile phone, a signal is transmitted to the wireless-transfer module within the gloves (Figure 6b), and the current intensity of the SMA is adjusted using the buttons in the application. When patients require gentle training, they can press the low-intensity mode on the application and the gloves will offer low force. If switched to high-intensity mode, the current intensity will increase, causing the SMA to generate a higher force. Patients and doctors can select the appropriate intensity mode from the three levels based on the patient's condition to achieve personalized rehabilitation training.

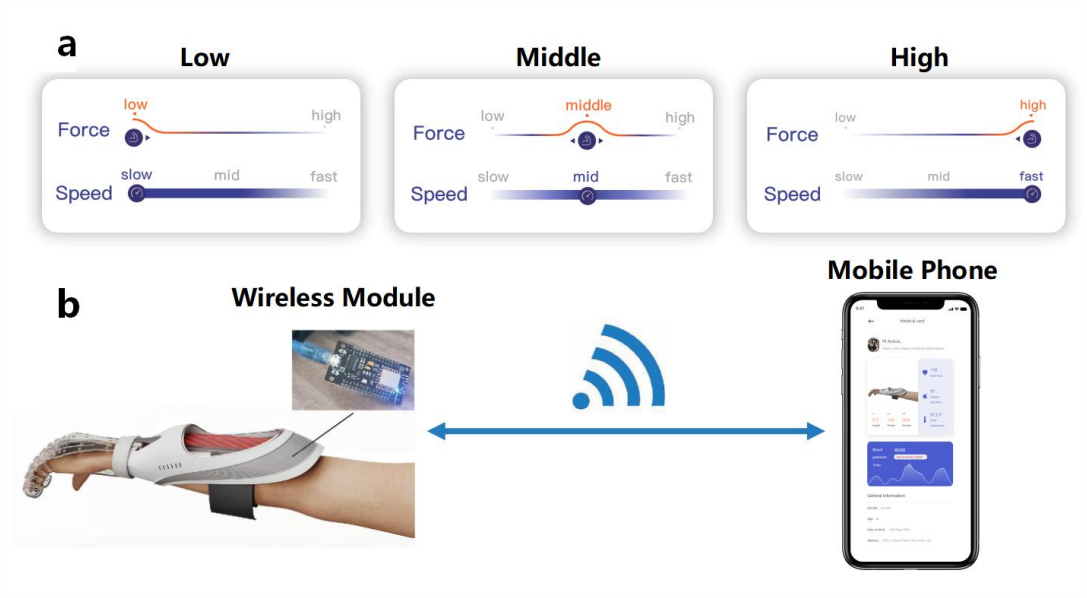

Figure 6. (**a**) Switch between different intensity modes. (**b**) Wireless interconnection of rehabilitation glove and mobile terminal.

4.3. Periodic Rehabilitation Training

To assess the feasibility of the amplifier structure, we sequentially used wires of different lengths on the structure for experimentation, as shown in Figure 7. The total length of the structure was 16 cm, corresponding to a single-fold wire length of 16 cm. By changing the number of winding turns, we achieved N-fold amplification, namely wires of 32 cm, 48 cm, 64 cm, 80 cm, and 96 cm. The displacement of the SMA wires at different amplification factors was measured using a scale, and it was found that the displacement was approximately proportional to the length, with a strain of around 5%. These results confirm the feasibility of the amplifier structure.

The wearing effect is shown in Figure 8a. To make the rehabilitation glove more portable, we use a small power-supply module (as shown in Figure 8b). The module has dimensions of 6 cm × 8 cm × 5 cm and consists of a switch button, a voltage adjustment button, a battery, and other components. By simply switching the voltage-adjustment button or using the mobile application, different voltage levels (0–12 V) can be supplied, enabling rehabilitation training modes of different intensities.

For the authenticity and effectiveness of the experiment, we used a silicone prosthetic hand model instead of a human hand for the rehabilitation training test. As shown in Figure 8c, the silicone prosthetic hand model was sequentially fitted with the rehabilitation glove, connected to the power-supply module, and paired with the mobile app via wireless transmission. First, we selected the low-intensity mode and pressed the start button. Then, the SMA wires embedded in the glove were heated up and began to contract. As the temperature increased, the contraction increased, and the fingers gradually extended. After 8 s, the middle finger reached the fully extended position. At this point, the current was cut off, and the SMA wires began to cool naturally through the ventilation openings. After 13 s, the glove returned to room temperature, and the fingers returned to their natural state. Similarly, we subsequently conducted rehabilitation training in the medium- and high-intensity modes. The results showed that with increasing intensity, the time required for rehabilitation training was significantly reduced. In the high-intensity mode, it took only 3 s to fully extend the fingers, achieving a finger bending angle change of approximately 34°, which can improve rehabilitation frequency, especially for patients with severe hand-function impairments.

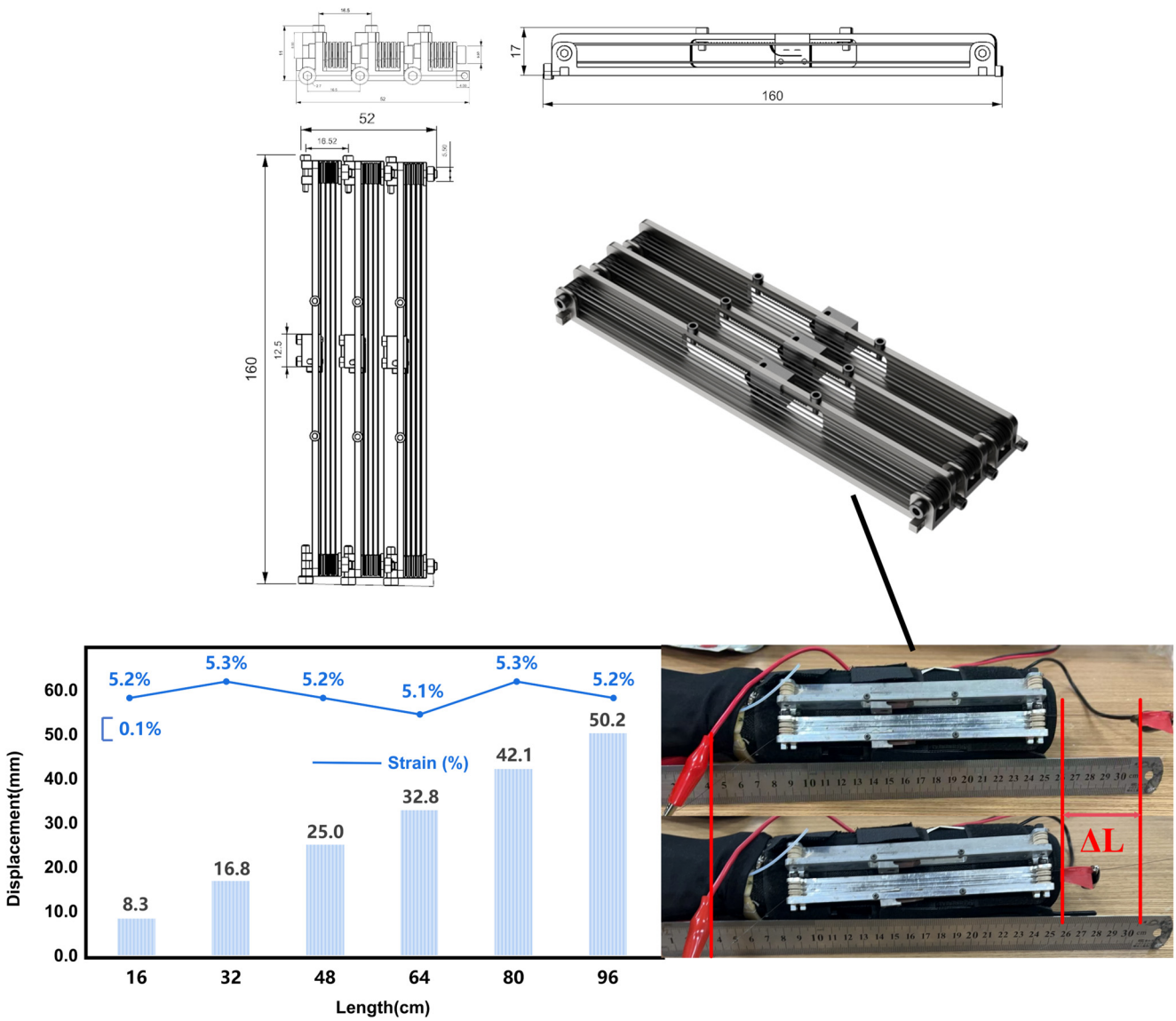

Figure 7. The relationship between wire length and displacement in the amplification structure.

To verify the technological advantages, we quantitatively compared our rehabilitation gloves with traditional electric and pneumatic gloves. Compared with Chen et al.'s motor-driven glove (350 g weight, 40 cm^3 volume) [16], our SMA-based prototype achieves a 50% weight reduction (180 g) and 40% volume shrinkage, primarily due to the amplifier structure that compresses the SMA wires from 680 mm to 160 mm. In contrast to Polygerinos et al.'s pneumatic glove, which requires 10 s for finger flexion and has a power density of 9 W/kg [17], our design demonstrates a 30% improvement in response time (7 s for 50 mm contraction) and 33% higher power density (12 W/kg). The mobile-app-integrated system further enables dynamic mode-switching, a feature absent in traditional fixed-training devices (e.g., Xie et al.) [6].

The development of more targeted and flexible training modes tailored to diverse patient needs remains at an early research stage. Future work will focus on creating freely adjustable modes (beyond existing preset options) to enable personalized therapy based on individual conditions and medical supervision. Further clinical trials are required to validate the proposed glove's practicality, including critical assessments of human–device compatibility (e.g., skin/joint interaction), long-term wearability, and kinematic synchronization.

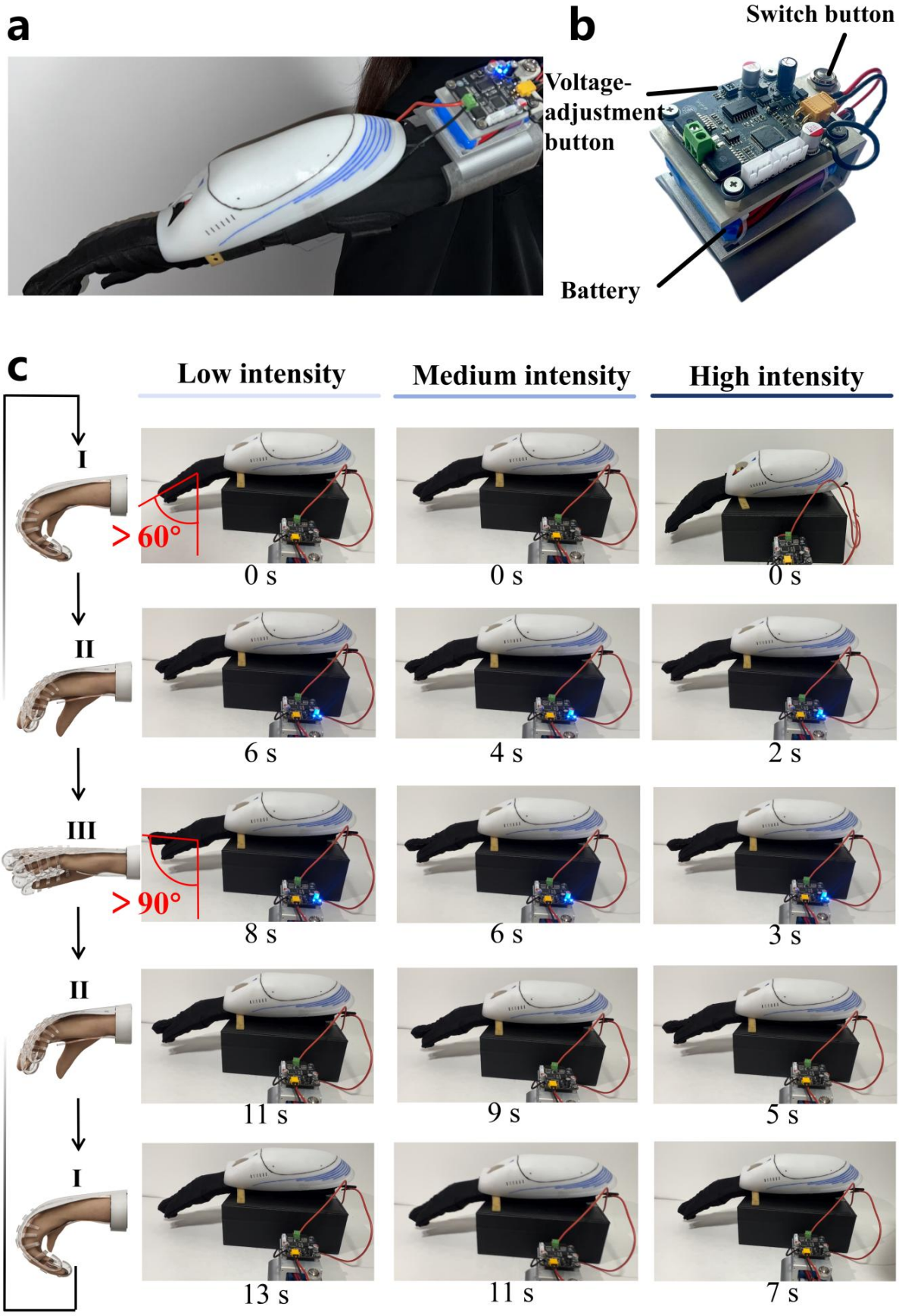

Figure 8. (**a**) Patient wearing rehabilitation glove. (**b**) Power-supply module. (**c**) Performance of rehabilitation exercises from (I) natural state to (II) intermediate state, and then to (III) fully extended state, under different training intensity modes.

5. Conclusions

This research harnessed the electro-thermomechanical coupling behavior of shape-memory alloys (SMAs) to develop a smart rehabilitation glove prototype for post-stroke hand-dysfunction recovery. The glove integrates an amplification mechanism, achieving compact dimensions and ultra-lightweight construction. Operating voltage requirements and actuation forces are governed by the phase-transformation kinetics of SMA wires. A companion mobile application enables wireless control with functionalities for patient health monitoring and dynamic adjustment of training intensity modes, facilitating personalized rehabilitation protocols and real-time clinician consultation. Experimental validation encompassed SMA mechanical characterization, wireless-module performance, and rehabilitation efficacy across intensity modes.

This smart rehabilitation glove features the following characteristics: (1) Superior portability: Unlike conventional motor-driven or pneumatic alternatives, the SMA-based system offers exceptional lightweight properties and wearability, enabling home-based and ambulatory rehabilitation. (2) Integrated telehealth: The mobile platform combines personalized health reporting, remote clinician consultation, and adjustable training modes, significantly reducing operational complexity and healthcare expenditures, particularly for elderly users. (3) Adaptive rehabilitation: Multi-level intensity modes (low/medium/high) permit patient-specific regimen adjustment according to clinical assessment, transforming monotonous exercise into engaging, individualized therapy.

6. Patents

The partial research results of this study have been patented (A Shape-Memory Alloy Rehabilitation Glove, Patent Authorization Number: CN 216,908,523 U).

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Abbreviations

The following abbreviations are used in this manuscript:

SMAs Shape-Memory Alloys

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